Understanding IPv4 Addressing and Subnetting

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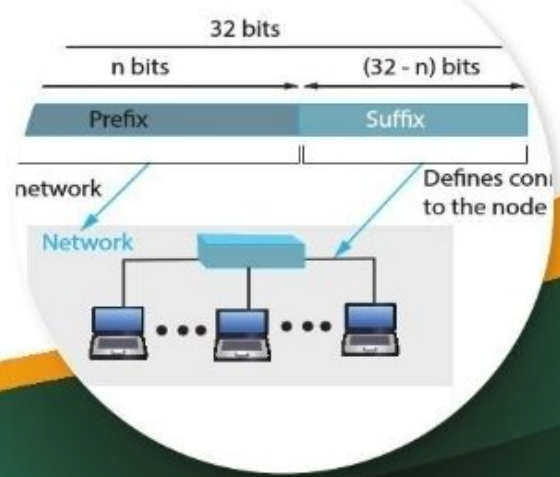
At the network layer, communication happens between host-to-host. There are many hosts available on the Internet. The hosts communicate with each other through the Internet. The data packet goes through several LANs or WANs before reaching the destination. To perform this kind of large-scale communication, a global addressing scheme is needed. This global addressing scheme, also called logical addressing, is known as IP addressing.

OBJECTIVES

By the end of this chapter, you will be able to:

- Understand the concept of IP addressing
- Describe address space allotted for classes of IPv4 address
- Describe subnetting and classless inter-domain routing
- Demonstrate subnetting in classes of IP addresses



Internet addresses are 32 bits in length and are referred to as IPv4 addresses or IP addresses. An IP address uniquely identifies a host or a router on the Internet. These addresses were classified according to their usage. But, when the Internet started expanding, there was a shortage of IP addresses. To resolve this problem, a short-term solution, called subnetting, was invented. Subnetting means to divide the IP addresses into subnets to create more networks.

This chapter provides information about classful addressing, subnetting, and how to create subnets from various classes of addresses.

IPv4 Addressing

CERTIFICATION OBJECTIVE

Understand the concept of IP addressing
Describe address space allotted for classes of IPv4 address

In the TCP/IP protocol, the IPv4 addresses are used to define the address of each node in the network. It is a numeric address used to identify each node uniquely on the network. It is a software address designed to allow nodes to communicate with each other regardless of their location and network.

IP Terminology

Some of the most commonly used terms in internet protocol addressing are explained below:

Bit: A bit consists of one digit. It can be either 1 or 0.

Nibble: A nibble consists of four digits.

Byte: A byte consists of 8 digits.

Octet: An octet is composed of 8 binary digit numbers. The term byte and octet are interchangeably used.

Network Address: A network address is used to send packets to the remote network using a routing scheme. 12.0.0.0 and 192.168.0.0 are examples of a network address.

Broadcast Address: A broadcast address is used to send information to all the nodes on a network. For example, 255.255.255.255 is used to send a packet on all nodes in any network; 192.168.255.255 is used to send a packet to all subnets and hosts on network 192.168.0.0; and 12.255.255.255 is used to send packets

to all subnets and hosts on network 10.0.0.0. Some of the common types of IP addresses used are shown in Table 1.

Table 1. Type of IP addresses

Type of IP Address	Definition
Reserved	Those addresses which cannot be assigned to any network device, such as the network address, the broadcast address, Class D, and Class E addresses.
Network ID	The network number assigned to a segment or VLAN.
Host ID	The address of a host in a network.
Directed Broadcast	The broadcast address of a network ID.
Local Broadcast	The address used to broadcast to all hosts on a network. For example, 255.255.255.255
Loopback	The internal address used for testing functions. For example, 127.0.0.1
Auto-configured	The address automatically assigned by a DHCP server.
Public	The address used to access nodes across any public network.
Private	The address used to access nodes in a private network. It cannot be used to access a public network.

The Hierarchical IP Addresses

The three common notations used to define an IPv4 address are binary notation (base 2), dotted-decimal notation (base 256), and hexadecimal notation (base 16).

An IPv4 address consists of 32 bits in binary notation. In the dotted-decimal notation, an IPv4 address is represented in decimal notation separated by dots. Each number ranges from 0 to 255. In network programming, the IPv4 addresses are used in hexadecimal notation. Each hexadecimal digit is equivalent to four bits. In a 32-bit address, there are 8 hexadecimal digits.

Figure 1 shows an IP address in the three notations.

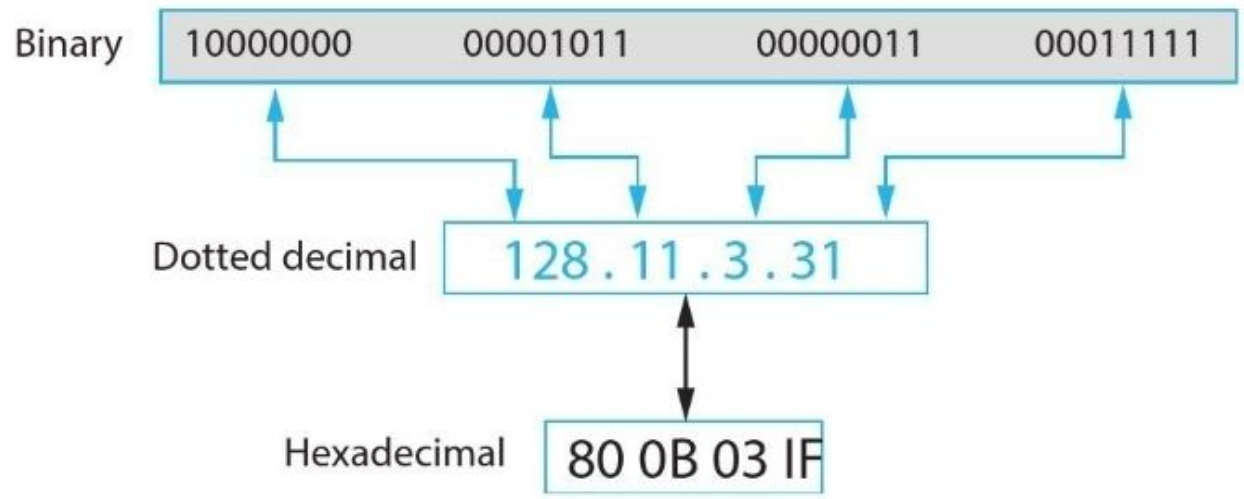


Figure 1: Notation in IPv4 addressing

A 32-bit IPv4 address is structured or hierarchical. It is divided into two parts. The prefix, which is the first part of the address, defines the network; the suffix, which is the second part of the address, defines the node. Figure 2 shows the prefix and suffix of a 32-bit IPv4 address. The prefix is of length *n* bits and the suffix is of length (32 – *n*) bits. The main advantage of using a hierarchical address is that it has a very large address space. Rather than using all the 32 bits as a unique identifier, the prefix is used to denote the network address and the suffix is used to denote either the subnet and host or just the node address.

IMPORTANT NOTES

In IPv4, the address space is 232or 4,294,967,296 (more than 4 billion).

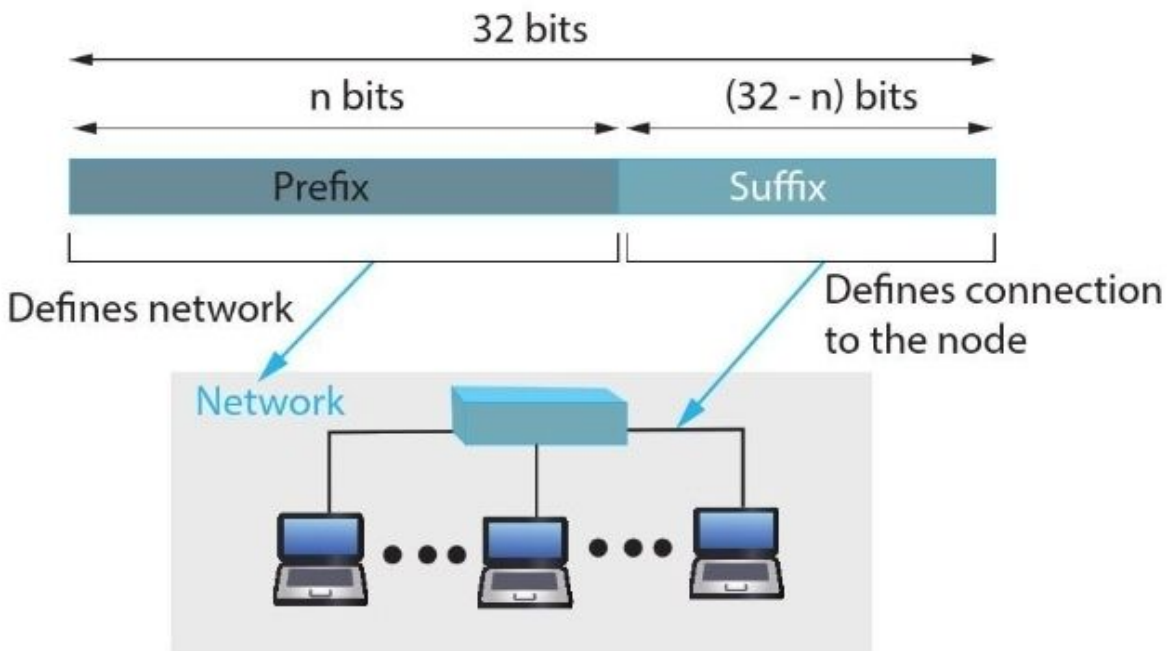


Figure 2: Hierarchy in addressing

Classful Addressing

In each network, there is a network address which uniquely identifies the network. Every node in the same network shares the same network address. The node address is assigned uniquely to every node. For example, in the IP address 172.168.25.26, 172.168 is the network address and 25.26 is the node address.

When the Internet was developed, the designers made classes based on network sizes. For a small number of networks having a large number of nodes, Class A addresses were allotted. On the other hand, Class C addresses were designed for a large number of networks having a small number of nodes. In between these classes, Class B address was designed. Figure 3 shows the classification of addresses.

	8 bits	8 bits	8 bits	8 bits
Class A:	Network	Host	Host	Host
Class B:	Network	Network	Host	Host
Class C:	Network	Network	Network	Host
Class D:	Multicast			
Class E:	Research			

Figure 3: Classification of addresses

Refer to Table 2 to learn more about the reserved IP addresses.

Table 2 Reserved IP addresses

Address	Function
Network address comprising all 0s	This address indicates 'this network or segment'.
Network address comprising all 1s	This address indicates 'all networks'.
Network address - 127.0.0.1	This address is reserved for loopback test. It is used to send a test packet to itself.
Node address comprising all 0s	This address refers to the network address of a specific network.
Node address comprising all 1s	This indicates 'all nodes' of a specific network.
Entire IP address comprising all 0s	This address is used by the Cisco routers to designate a default route.
Entire IP address comprising all 1s	This address is used to broadcast to all nodes in the current network.

Class A Addresses

In the Class A address, the first octet represents the network address and the three remaining octets represent the node addresses. The Class A format is:

network. node. node. node

In the Class A address, the first bit of the first octet must always be 0 which implies that only seven bits are used as network identifiers. This means that a Class A address has only 128 (2^7) networks. The range of first octet of a Class A network address is:

00000000 = 0

01111111 = 127

In Class A IP addresses, the range of valid IP addresses is 0.0.0.0 to 127.255.255.255. For example, in the IP address 116.36.125.96, 116 is the network address and 36.125.96 is the node address. Every node on this network will have the network address of 116.

The network address of all 0s, 1s and the address 127 is reserved, therefore, only numbers 1 to 126 are used to designate Class A network addresses. This means that the actual number of usable Class A network addresses is 128 minus 2, or 126. Each Class A address has 3 octets (24-bits) for the node address of a machine. There are 2^{24} or 16,777,216 possible unique node addresses. Since node addresses with all 0s and 1s are reserved, therefore, the actual maximum usable number of nodes for a Class A network is 2^{24} - 2, which equals 16,777,214.

Class B Address

In a Class B address, the first two octets represent the network address and the last two octets represent the node addresses. The Class B format is:

network. network. node. node

In a Class B address, the first bit of the first octet must always be 1 and the second bit must be 0. This means that a Class B address can have only 16,384 (2^{14}) networks. The range of first octet of a Class B network address is:

10000000 = 128

10111111 = 191

In Class B IP addresses, the range of valid IP addresses is 128.0.0.0 to 191.255.255.255. For example, in the IP address 129.36.125.96, 129.36 represents the network address and 125.96 represents the node address. Every node on this network will have a network address of 129.36.

A Class B address uses 2 octets for node addresses. This means, there are 2^{16} minus the two reserved patterns (all 0s and all 1s), a total of 65,534 possible node addresses.

Class C Address

In a Class C address, the first three octets are assigned to the network address and the last remaining octet shows the node address. The Class C format is:

network. network. network. node

In a Class C address, the first two bits of first octet must always be 1 and the third bit must be 0. This means that a Class C address can have only 2,097,152 (2^{21}) networks. The range of first octet of a Class C network address is:

11000000 = 192

11011111 = 223

In Class C IP addresses, the range of valid IP addresses is 192.0.0.0 to 223.255.255.255. A Class C address uses 1 octet for node addresses. This means that there are (2^8 – (two reserved patterns)), that is, a total of 254 possible node addresses. This address consists of all 0s and all 1s.

Class D Address

In a Class D address, the first three bits of first octet must always be 1 and the fourth bit must be 0. These addresses are reserved for multicasting and there is no network address or host address. The range of first octet of a Class D network address is:

11100000 = 224

11101111 = 239

In Class D IP addresses, the range of valid IP addresses is 224.0.0.0 – 239.255.255.255.

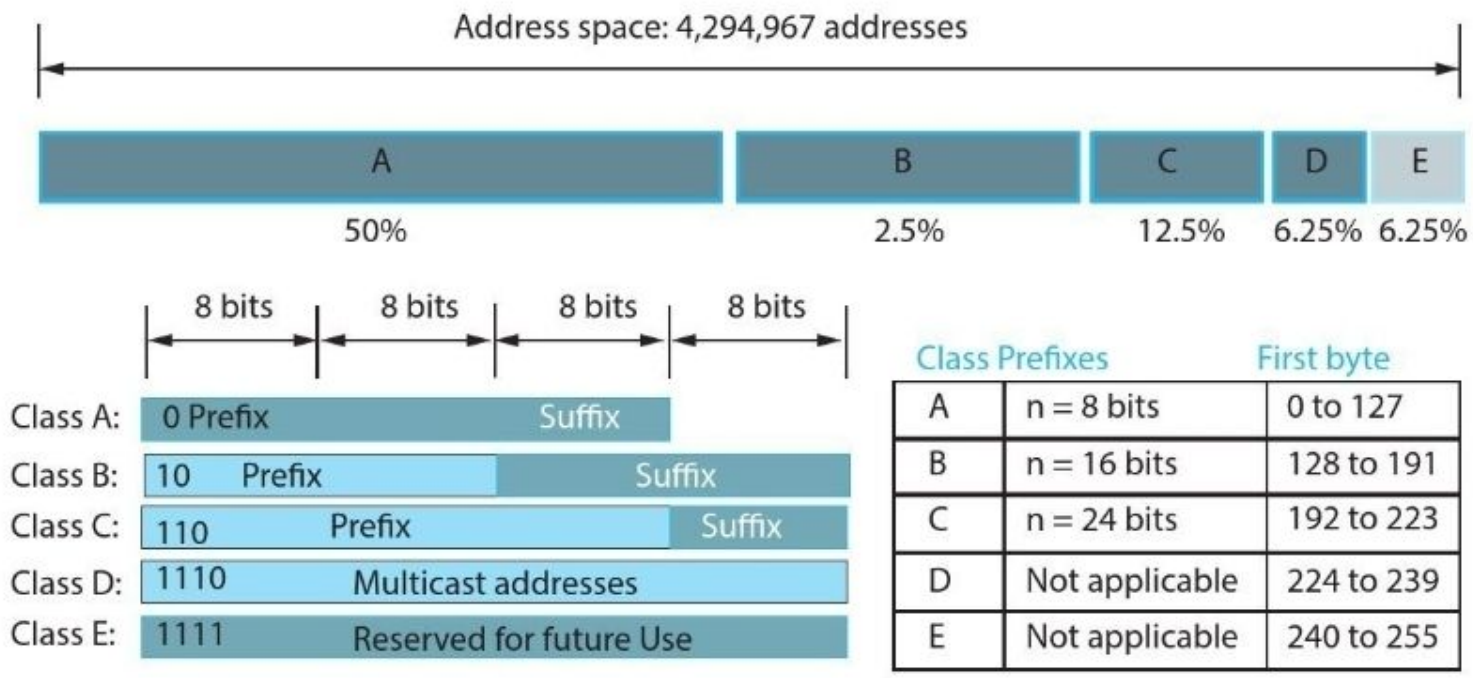
Class E Address

In a Class E address, the first four bits of first octet must always be 1. These addresses are reserved for scientific purposes and future use. There is no network address or host address in a Class E address. The range of first octet of a Class E network address is:

11110000 = 240

11111111 = 255

In Class E IP addresses, the range of valid IP addresses is 240.0.0.0 – 255.255.255.255. Figure 4 shows the address space in classful addressing.



Class Prefixes		First byte
A	n = 8 bits	0 to 127
B	n = 16 bits	128 to 191
C	n = 24 bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255

Figure 4: Summary of addresses

Private IP Addresses

A private IP address is an address that is assigned to the computers by a router or other Network Address Translation (NAT) device. These addresses are reserved for private use. A private IP address is also known as local IP address. A public IP address can be accessed over the Internet, but a private IP address cannot be accessed directly over the Internet.

In RFC 1919, the private address space is defined by the following blocks:

10.0.0.0/8: It is a Class A network which allows the following range of valid IP addresses- 10.0.0.1 to 10.255.255.254. This network has 24 host bits.

172.16.0.0/12: It is a Class B network which allows the following range of IP addresses- 172.16.0.1 to 172.31.255.254. This network has 20 host bits.

192.168.0.0/16: It is a Class C network which allows the following range of IP addresses- 192.168.0.1 to 122.168.255.254. This network has 16 host bits.

ANSWER IT

1. 26.8.16.198 IP address belongs to
   a. Class A address
   b. Class B address
   c. Class C address
   d. Class D address
2. Which class of address is reserved for multicast addresses?
   a. Class A address
   b. Class B address
   c. Class C address
   d. Class D address
3. Which of the following are valid Class C IP addresses that can be assigned to hosts?
   a. 1.1.1.1
   b. 200.1.1.1
   c. 128.128.128.128
   d. 224.1.1.1
4. How many IP addresses does each Class B network contain that can be assigned to hosts?
   a. 16,777,214
   b. 65,536
   c. 65,534
   d. 65,532
5. The format of Class C address is
   a. Network. Node. Node. Node
   b. Node. Node. Node. Node
   c. Network. Network. Network. Node
   d. Network. Network. Node. Node

Subnetting Basics

CERTIFICATION OBJECTIVE

Describe subnetting and classless inter-domain routing
Demonstrate subnetting in classes of IP addresses

With the growth of the Internet, there was a need for larger address spaces. This could only be achieved by increasing the length of the IP addresses. But, this meant that the format of the IP address needed to be changed. For this, a short-term solution known as classless addressing was provided which used subnetting. Subnetting refers to a logical division of an IP address.

Advantages of Subnetting

Some of the advantages of subnetting are:

Reduced network traffic: The packets which are destined for other networks are routed through routers. Routers create broadcast domains. The more broadcast domains you create, the smaller they are and the less network traffic on each network segment there is.

Optimised network performance: Reduced network traffic results in optimised network performance.

Simplified management: Small broadcast domains are simpler to manage.

Facilitated spanning of large geographical distances: Small networks connected using LAN are more efficient than the WAN links because WAN links are considerably slower and more expensive.

Subnet Masks

A part of the host address is used as the subnet address. This is done by assigning a subnet mask. A subnet mask is a 32-bit value which allows the receiving node to distinguish between the network address and the host address. A 32-bit subnet mask is composed of 1s and 0s. The 1s refer to the network or subnet. The default subnet masks for the three different classes of IP addresses are:Class A has subnet mask of 255.0.0.0

- Class B has subnet mask of 255.255.0.0
- Class C has subnet mask of 255.255.255.0

Classless Inter-Domain Routing (CIDR)

The prefix indicates the network address. Hence, in the classless addressing, the prefix length, n, is added to the address separated by a slash. This notation is referred to as classless inter-domain routing (CIDR). An address in classless addressing is shown in Figure 5.

IMPORTANT NOTES

The best subnet mask for a point-to-point link, carrying only two host addresses, are needed, is 255.255.255.252.

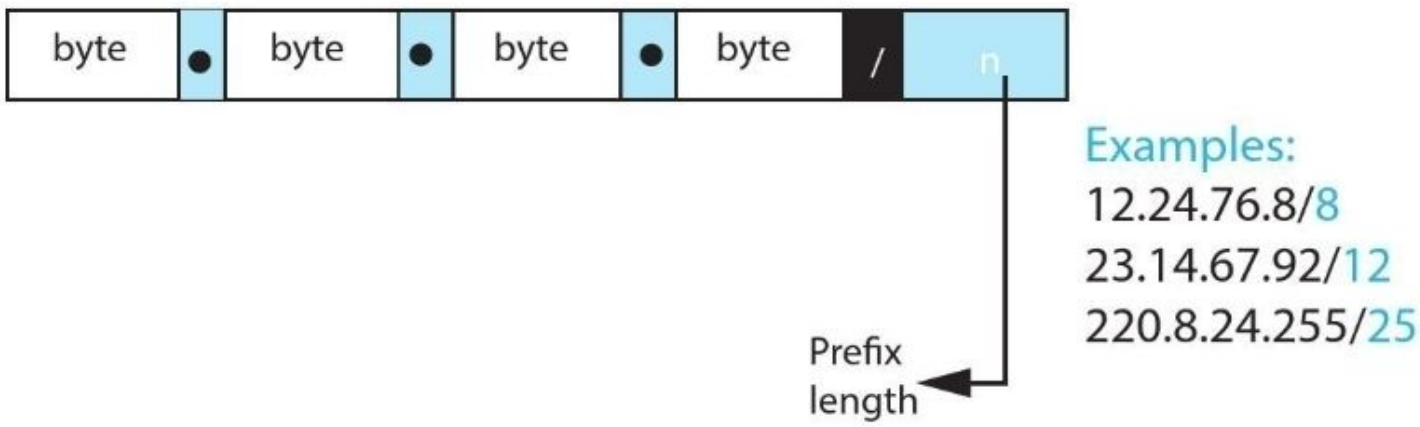


Figure 5: Summary of addresses

Table 3 lists all available subnet masks and their equivalent CIDR slash notation.

Table 3 CIDR Values

Subnet Mask	CIDR Value
255.0.0.0	/8
255.128.0.0	/9
255.192.0.0	/10
255.224.0.0	/11
255.240.0.0	/12
255.248.0.0	/13
255.252.0.0	/14
255.254.0.0	/15
255.255.0.0	/16
255.255.128.0	/17
255.255.192.0	/18
255.255.224.0	/19
255.255.240.0	/20
255.255.248.0	/21
255.255.252.0	/22
255.255.254.0	/23
255.255.255.0	/24
255.255.255.128	/25
255.255.255.192	/26
255.255.255.224	/27
255.255.255.240	/28
255.255.255.248	/29
255.255.255.252	/30

Subnetting Class C Addresses

In a Class C address, only 8 bits are available for the host addresses. The Class C subnet masks are:

```
Binary   Decimal CIDR
--------------------
00000000  = 0    /24
10000000  = 128   /25
11000000  = 192   /26
11100000  = 224   /27
11110000  = 240   /28
11111000  = 248   /29
11111100  = 252   /30
```

You cannot select CIDR /31 and /32 because there should be at least 2 hosts bits available for assigning IP addresses to the hosts.

While choosing a subnet mask for your network, you need to answer five questions:

1. How many subnets does the chosen subnet mask produce?
2. How many valid hosts per subnet are available?
3. What are the valid subnets?
4. What is the broadcast address of each subnet?
5. What are the valid hosts in each subnet?

Answers to the above questions can be found in the following ways:

1. How many subnets?

   Number of subnets = 2^x. X defines the number of masked bits or the 1s. For example, in 11100000, there are $2^3 = 8$ subnets.

2. How many hosts per subnet?

   Number of hosts per subnet = $2^y - 2$. Y defines the number of unmasked bits or the 0s. For example, in 11100000, the number of 0s gives us $2^5 - 2 = 30$ hosts.

3. What are valid subnets?

   Block Size = 256 – subnet mask. For example, if a subnet mask has CIDR value of 224, then the total number of valid subnets would be 256 – 224 = 32. The block size of a 224 mask will always be 32.

4. What is the broadcast address for each subnet?

   Since the counted subnets would be 0, 32, 64, 96, 128, 160, 192, and 224, the broadcast address will always be the number preceding the next subnet. For example, the broadcast address of 0 subnet would be 31 because the next subnet is 32. Similarly, the broadcast address of 32 subnet would be 63 because the next subnet is 64 and the broadcast address of the last subnet will always be 255.

5. What are valid hosts?

   Valid hosts are the numbers between the subnets, omitting the all-0s and all-1s. For example, if 32 is the subnet number and 127 is the broadcast address, then 32–126 is the valid host range.

Now, we will discuss examples of how to find a Class C subnet mask.

Example 1: 255.255.255.128 (/25)

128 is represented as 10000000 in binary, therefore, there is only 1 bit for subnetting and 7 bits for hosts. Let us subnet the Class C network address 192.168.20.0.

192.168.20.0 = Network address

255.255.255.128 = Subnet mask

1. How many subnets?

   There is only one 1 bit, so, there are $2^1 = 2$ subnets.

2. How many hosts per subnet?

   There are seven 0 bits, so, there are $2^7 - 2 = 126$ hosts.

3. What are the valid subnets?

   256 – 128 = 128. The subnets are 0, 128.

4. What's the broadcast address for each subnet?

   The broadcast of the 0 subnet is 127. The broadcast of 128 subnet is 255

5. What are the valid hosts?

Subnet	0	128
First Host	192.168.20.1	192.168.20.129
Last Host	192.168.20.126	192.168.20.254

Example 2: 255.255.255.192 (/26)

192 is represented as 11000000 in binary, therefore, there are two 1s for subnetting and six 0s for hosts. Let us subnet the Class C network address 192.168.20.0.

192.168.20.0 = Network address

255.255.255.192 = Subnet mask

1. How many subnets?

   There are two 1s, so, there are $2^2 = 4$ subnets.

2. How many hosts per subnet?

   There are six 0s, so, there are $2^6 - 2 = 62$ hosts.

3. What are the valid subnets?

   256 – 192 = 64. The subnets are 0, 64, 128, and 192.

4. What's the broadcast address for each subnet?

   The broadcast address of each subnet is:

- 0 subnet -> broadcast address 63
- 64 subnet -> broadcast address 127
- 128 subnet -> broadcast address 191.
- 192 subnet -> broadcast address 255.

5. What are the valid hosts?

Subnet	0	64	128	192
First Host	192.168.20.1	192.168.20.65	192.168.20.129	192.168.20.193
Last Host	192.168.20.62	192.168.20.126	192.168.20.190	192.168.20.254

With the help of above examples, Table 4 shows you the information about the Class C subnet masks.

Table 4 Class C Subnetting summary

CIDR Notation	Mask	Bits	Block Sizes	Subnets	Hosts
/25	128	1 bit 1s and 7 bits 0s	128	0, 128	2 subnets, each with 126 hosts
/26	192	2 bit 1s and 6 bits 0s	64	0, 64, 128, 192	4 subnets, each with 62 hosts
/27	224	3 bit 1s and 5 bits 0s	32	0, 32, 64, 96, 128, 160, 192, 224	8 subnets, each with 30 hosts
/28	240	4 bit 1s and 4 bits 0s	16	0, 16, 32, 48, etc.	16 subnets, each with 14 hosts
/29	248	5 bit 1s and 3 bits 0s	8	0, 8, 16, 24, etc.	32 subnets, each with 6 hosts
/30	252	6 bit 1s and 2 bits 0s	4	0, 4, 8, 16, 20, etc.	64 subnets, each with 2 hosts

Subnetting Class B Addresses

In a Class B address, only 16 bits are available for the host addresses. The Class B subnet masks are:

```
Binary                Decimal   CIDR
-----------------------------------
00000000.00000000  = 0.0       /16
10000000.00000000  = 128.0     /17
11000000.00000000  = 192.0     /18
11100000.00000000  = 224.0     /19
11110000.00000000  = 240.0     /20
11111000.00000000  = 248.0     /21
11111100.00000000  = 252.0     /22
11111110.00000000  = 254.0     /23
11111111.00000000  = 255.0     /24
11111111.10000000  = 255.128   /25
11111111.11000000  = 255.192   /26
11111111.11100000  = 255.224   /27
11111111.11110000  = 255.240   /28
11111111.11111000  = 255.248   /29
11111111.11111100  = 255.252   /30
```

The following examples demonstrate how to subnet with Class B address just like in Class C:

Example 1: 255.255.128.0 (/17)

128 is represented as 10000000 in binary, therefore, there is only 1 bit for subnetting and 15 bits for hosts.

172.30.0.0 = Network address

255.255.128.0 = Subnet mask

1. How many subnets?

   There is only one 1 bit, so, there are $2^1 = 2$ subnets.

2. How many hosts per subnet?

   There are fifteen 0 bits (seven in first octet and eight in second), so, there are $2^{15} - 2 = 32{,}766$ hosts.

3. What are the valid subnets?

   Block Size = 256 – subnet mask

   256 – 128 = 128The subnets are 0.0, 128.0

4. What's the broadcast address for each subnet?

   The broadcast of the 0.0 subnet is 127.255 and the broadcast of the 128.0 subnet is 255.255

5. What are the valid hosts?

Subnet	0.0	128.0
First Host	172.30.0.1	172.30.128.1
Last Host	172.30.127.254	172.30.255.254

Example 2: 255.255.192.0 (/18)

192 is represented as 11000000 in binary, therefore, there are two 1s for subnetting and fourteen 0s for hosts.

172.30.0.0 = Network address

255.255.192.0 = Subnet mask

1. How many subnets?

   There are two 1s, so, there are $2^2 = 4$ subnets.

2. How many hosts per subnet?

   There are fourteen 0s, so, there are $2^{14} – 2 = 16{,}382$ hosts.

3. What are the valid subnets?

   Block Size = 256 – subnet mask

   256 – 192 = 64The subnets are 0.0, 64.0, 128.0, 192.0

4. What is the broadcast address for each subnet?

   The broadcast address of each subnet is:

- 0.0 subnet -> broadcast address 63.255
- 64.0 subnet -> broadcast address 127.255
- 128.0 subnet -> broadcast address 191.255
- 192.0 subnet -> broadcast address 255.255.

5. What are the valid hosts?

Subnet	0.0	64.0	128.0	192.0
First Host	172.30.0.1	172.30.64.1	172.30.128.1	172.30.192.1
Last Host	172.30.63.254	172.30.127.254	172.30.191.254	172.30.255.254

With the help of the above examples, Table 5 lists the information about the Class B subnet masks.

CIDR Notation	Mask	Bits	Subnets	Hosts
/17	128.0	1 bit 1s and 15 bits 0s	2	Each with 32,766 hosts
/18	192.0	2 bit 1s and 14 bits 0s	4	Each with 16,382 hosts
/19	224.0	3 bit 1s and 13 bits 0s	8	Each with 8,190 hosts
/20	240.0	4 bit 1s and 12 bits 0s	16	Each with 4,094 hosts
/21	248.0	5 bit 1s and 11 bits 0s	32	Each with 2,046 hosts
/22	252.0	6 bit 1s and 10 bits 0s	64	Each with 1,022 hosts
/23	254.0	7 bit 1s and 9 bits 0s	128	Each with 510 hosts
/24	255.0	8 bit 1s and 8 bits 0s	256	Each with 254 hosts
/25	255.128	9 bit 1s and 7 bits 0s	512	Each with 126 hosts
/26	255.192	10 bit 1s and 6 bits 0s	1024	Each with 62 hosts
/27	255.224	11 bit 1s and 5 bits 0s	2048	Each with 30 hosts
/28	255.240	12 bit 1s and 4 bits 0s	4096	Each with 14 hosts
/29	255.248	13 bit 1s and 3 bits 0s	8192	Each with 6 hosts
/30	255.252	14 bit 1s and 2 bits 0s	16384	Each with 2 hosts

Subnetting Class A Addresses

In a Class A address, 24 bits are available for the host addresses. The Class A subnet masks are:

```
Binary                          Decimal      CIDR
-----------------------------------------------
00000000.00000000.00000000  = 0.0.0        /8
10000000.00000000.00000000  = 128.0.0      /9
11000000.00000000.00000000  = 192.0.0      /10
11100000.00000000.00000000  = 224.0.0      /11
11110000.00000000.00000000  = 240.0.0      /12
11111000.00000000.00000000  = 248.0.0      /13
11111100.00000000.00000000  = 252.0.0      /14
11111110.00000000.00000000  = 254.0.0      /15
11111111.00000000.00000000  = 255.0.0      /16
11111111.10000000.00000000  = 255.128.0    /17
11111111.11000000.00000000  = 255.192.0    /18
11111111.11100000.00000000  = 255.224.0    /19
11111111.11110000.00000000  = 255.240.0    /20
11111111.11111000.00000000  = 255.248.0    /21
11111111.11111100.00000000  = 255.252.0    /22
11111111.11111110.00000000  = 255.254.0    /23
11111111.11111111.00000000  = 255.255.0    /24
11111111.11111111.10000000  = 255.255.128  /25
11111111.11111111.11000000  = 255.255.192  /26
11111111.11111111.11100000  = 255.255.224  /27
11111111.11111111.11110000  = 255.255.240  /28
11111111.11111111.11111000  = 255.255.248  /29
11111111.11111111.11111100  = 255.252.252  /30
```

The following example shows you how to subnet with a Class A address just like Class B and C:

Example 1: 255.255.240.0 (/20)

240 is represented as 11110000 in binary, therefore, there are twelve 1s for subnetting (eight 1s from second octet and four 1s from third octet) and twelve 0s for hosts.

1. How many subnets?

   $2^{12} = 4096$ subnets.

2. How many hosts per subnet?

   $2^{12} - 2 = 4094$ hosts.

3. What are the valid subnets?

   Block Size = 256 – subnet mask

   256 – 240 = 16

   The subnets are 0.0.0, 0.16, 0.32, 0.0.64, …, 255.240.0

4. What is the broadcast address for each subnet?

   The broadcast address of each subnet is:

- 0.0.0 subnet -> broadcast address 0.15.255
- 0.16.0 subnet -> broadcast address 0.127.255
- 0.128.0 subnet -> broadcast address 0.191.255
- 0.192.0 subnet -> broadcast address 0.255.255.
- …
- 255.240.0 subnet -> broadcast address 255.255.255

5. What are the valid hosts?

Subnet	0.0.0	0.16.0	0.32.0	…	255.240.0
First Host	10.0.0.1	10.0.16.1	10.0.32.1	…	10.255.240.1
Last Host	10.0.15.254	10.0.31.254	10.0.47.254	…	10.255.255.254

Recap

- IP addresses are 32 bits in length and are broken into 4 octets of 8 bits separated by a dot. This notation is called dotted-decimal.
- IP addresses are divided into five classes: A (1–126), B (128–191), C (192–223), D (224–239), and E (240–254).
- IP addresses have two parts: network and host. Class A address has the first octet, Class B has first 2 octets, and Class C has first 3 octets as network ID.
- IP addresses can be identified by the first few bits of the first octet. Class A addresses begin with 0, Class B with 10, Class C with 110, Class D with 1110, and Class E with 11110.
- In subnetting, IP addresses are broken and used efficiently. Bits from the host ID are used to create more subnets with smaller number of hosts.
- Subnet masks are 32 bits long and are represented as a dotted-decimal (such as 255.255.255.0) or by using slash notation such as /17.

Test Your Skills

Multiple Choice Questions

1. What is the address range of a Class B network address in binary?
   a. 01xxxxxx
   b. 0xxxxxxx
   c. 10xxxxxx
   d. 110xxxxx
2. Which class of IP address provides a maximum of only 254 host addresses per network ID?
   a. Class A
   b. Class B

c. Class C

d. Class D

3. In a Class C network, there are 25 bits for networking. How many subnets will that Class C address have?

   a. 1

   b. 2

   c. 3

   d. 4

4. Which of the following is a valid subnet mask value?

   a. 255.0.255.255

   b. 0.0.0.255

   c. 255.255.254.0

   d. 255.255.255.256

5. What is the maximum number of IP addresses that can be assigned to hosts on a local subnet that uses the 255.255.255.240 subnet mask?

   a. 14

   b. 15

   c. 16

   d. 30

6. If an Ethernet port on a router was assigned an IP address of 172.16.102.21/25, what would be the valid subnet address of this host?

   a. 172.16.112.0

   b. 172.16.0.0

   c. 172.16.96.0

   d. 172.16.255.0

7. The network address of 192.168.0.0/19 provides how many subnets and hosts?
   a. 7 subnets, 30 hosts each
   b. 8 subnets, 8,190 hosts each
   c. 8 subnets, 2,046 hosts each
   d. 7 subnets, 2,046 hosts each
8. To test the IP stack on your local host, which IP address would you ping?
   a. 127.0.0.0
   b. 1.0.0.127
   c. 127.0.0.1
   d. 127.0.0.255
9. What is the subnetwork address for a host with the IP address 198.40.7.65/28?
   a. 198.40.7.56
   b. 198.40.7.32
   c. 198.40.7.64
   d. 198.40.7.0
10. Which of the following is a Class C IP address?
   a. 10.10.14.118
   b. 135.23.112.57
   c. 191.200.199.199
   d. 204.67.118.54

Scenario Based Questions

1. Your router has the following IP address on Ethernet0: 172.16.2.1/23. Which of the following can be valid host IDs on the LAN interface attached to the router?
   i. 172.16.1.100
   ii. 172.16.1.198
   iii. 172.16.2.255
   iv. 172.16.3.0
   a. i only
   b. ii and iii only
   c. iii and iv only
   d. None of the above
2. You need to subnet a network that has 5 subnets, each with at least 16 hosts. Which classful subnet mask would you use?
   a. 255.255.255.192
   b. 255.255.255.224
   c. 255.255.255.240
   d. 255.255.255.248
3. You have an interface on a router with the IP address of 192.168.145.15/29. Including the router interface, how many hosts can have IP addresses on the LAN attached to the router interface?
   a. 6
   b. 8
   c. 30
   d. 32

4. You need to configure a server that is on the subnet 172.16.19.24/29. The router has the first available host address. Which of the following should you assign to the server?
   a. 172.16.19.0 255.255.255.0
   b. 172.16.19.33 255.255.255.240
   c. 172.16.19.26 255.255.255.248
   d. 172.16.19.31 255.255.255.248
5. You have a network with a subnet of 172.16.17.0/22. Which is the valid host address?
   a. 172.16.17.1 255.255.255.252
   b. 172.16.0.1 255.255.240.0
   c. 172.16.20.1 255.255.254.0
   d. 172.16.18.255 255.255.252.0